



Qualitative analysis of healthcare provider perspectives to evaluating beta-lactam allergies

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SUMMARY

Background: There is a lack of understanding of the barriers reported by healthcare providers when evaluating beta-lactam allergies, but knowledge of these barriers is required for practical and effective implementation interventions.

Methods: Twenty-five healthcare providers, consisting of physicians, nurses and pharmacists practicing in the areas of intensive care, emergency medicine, infectious disease and general hospital practice, were interviewed between September 2021 and July 2023. Twenty-three of these providers were practising in the USA. A semi-structured interview guide grounded in the Theoretical Domain Framework was used for the interviews. Deductive and inductive analysis was performed on the interview transcripts, and translated into intervention recommendations using the Behaviour Change Wheel.

Results: Widely held beliefs included a lack of clear policy for the evaluation of allergies, confusing or missing documentation of allergy information, confidence in their own and their colleagues' ability to evaluate allergies when information is available, and pharmacists as the provider most equipped to evaluate beta-lactam allergies.

Conclusions: Health systems should adopt and disseminate policies for the evaluation of beta-lactam allergies, and promote the use of pharmacists in the evaluation of drug allergies when possible. Allergy sections of electronic health records should be reworked to encourage unambiguous documentation of allergy reactions and support using previously tolerated beta-lactam antibiotics.

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Introduction

Beta-lactam (BL) antibiotics, which include penicillins, cephalosporins and carbapenems, are the most widely used antibiotics in the world [1]. Unfortunately, >5–13% of the population has an allergy to at least one BL-containing product listed in their electronic health record (EHR) [2,3]. These

allergies limit the ability of clinicians to utilize BL antibiotics and result in increased reliance on second-line antibiotics; this is associated with adverse health outcomes [4–7]. However, much of this clinical detriment is potentially avoidable as <10% of listed BL allergies represent true immune-mediated reactions, and many patients with listed BL allergies can tolerate many BL-containing products if challenged [8–10].

There have been recent advances in the understanding of BL allergies, and tools such as risk stratification methods and decision support alerts have been created to assist in evaluating the legitimacy of reported BL allergies [11–15]. Despite these tools, it has been shown that more than five new allergies

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are entered for every one that is deleted in US-based EHRs [16]. Physicians remain hesitant to challenge BL allergies despite reporting a high level of awareness of the rates of erroneous allergies in practice [17,18]. The stagnation in evaluation of BL allergies in the setting of increased understanding of evaluation methods suggests that there are existing barriers in practice that are limiting the uptake and usage of BL-allergy evaluation and delabelling techniques. Barriers to the evaluation techniques need to be further understood for interventions and tools to have the intended effects in reducing erroneous BL allergies.

Qualitative studies interviewing healthcare providers on the evaluation of BL allergies are well suited to analysing the disconnect that is limiting the progress of BL-allergy evaluation and delabelling. Two previous UK-based studies conducted interviews and focus groups with primary care providers, pharmacists and nurses, and patients found that providers were not aware of evaluation resources at their hospital, felt that the evaluation of allergies is a complex task, and felt that the evaluation of BL allergies is not a high priority in their care setting relative to other tasks; however, these studies only included a limited number of specialists, and it is unknown how their results may compare with care settings outside the UK [19,20]. It has not been well described whether similar barriers are being experienced outside the UK, and it is important to evaluate differing viewpoints to generate consensus on the most important barriers which are being experienced and are hindering BL-allergy evaluation and delabelling.

As such, this study sought to address the current limitations using semi-structured interviews with a diverse group of largely North American-based clinicians, including pharmacists, nurses and physicians. Additionally, beliefs may differ considerably based on clinical specialty and setting, so this study attempted to include these differing viewpoints by including specialists from the fields of intensive care, emergency medicine and infectious disease in addition to primary and general practice. Finally, the interviews were aligned with a standardized framework for behaviour change to translate the interview results systematically into actionable evidence-based implementation interventions that are designed to overcome the reported barriers that are occurring in practice [21].

Methods

Study design

This qualitative study utilized semi-structured interviews to evaluate the current barriers to BL-allergy evaluation and delabelling. The process outlined by Michie *et al.* in the Behaviour Change Wheel (BCW), which utilizes the Theoretical Domains Framework (TDF) and the Capability, Opportunity and Motivation-Behaviour (COM-B) Model, was used to relate behaviour domains to proposed actionable interventions. The study was conducted in accordance with the standards for reporting in qualitative research, and was approved as exempt by the University of Pittsburgh International Review Board [22].

Framework

The TDF was used to structure the barrier assessment for the evaluation of BL allergies. The TDF is the consensus of 33

behaviour change models and consists of 14 domains which influence behaviours such as reinforcement, skills and knowledge [23]. The TDF domains are then further divided into 84 constructs that describe feelings and motivations. For example, both rewards and punishments are constructs contained within the reinforcement domain.

The COM-B/BCW contains direct links between the TDF domains and respective intervention functions, allowing for systematic mapping of identified beliefs into intervention categories following an eight-step process [24]. Steps 1–3 are designed to define target behaviour clearly. In the present study, this was completed through prior knowledge and literature evaluation [21,23]. The target behaviour identified for evaluation was 'healthcare providers evaluating the legitimacy of beta-lactam allergies listed in electronic medical records'. Steps 4–6 are designed to understand what needs to change, and assess the nine possible intervention functions such as persuasion, training and restriction. In the present study, this was determined by stakeholder input through the interview process described below. Steps 7 and 8 are used to translate the intervention functions into pragmatic individual behaviour change techniques.

Participants

This study used a convenience sampling approach of healthcare providers well known to the study team, followed by snowball recruitment of additional providers [25,26]. Participants were contacted initially through e-mail, and were further contacted to schedule an interview date if they agreed to participate. No compensation was received for participation. Prior knowledge of the evaluation of BL allergies was not a factor in recruitment, which was based solely on willingness and desire to participate.

Data collection

Semi-structured interviews were used for data collection because they can be adapted to a group of interviewees with differing perceptions and backgrounds, which was an expectation of the interview cohort consisting of 11 combinations of healthcare providers and clinical specialties [27]. A preliminary interview guide was developed using background knowledge on the target behaviour, and contained questions designed specifically to align with domains and constructs in the TDF [28]. The interview guide was field tested with a pharmacist who is experienced in conducting qualitative research, and further refined with probing questions and improved clarity [29]. The revised interview guide was used for all interviews (Appendix 1, see online supplementary material).

Interviews were conducted between September 2021 and July 2023. Interviews took place online and were audio recorded using Zoom videoconferencing software [30]. Interviewees verbally consented to be recorded prior to the interview, and the interviews lasted for 30–60 min. The recordings were then fully transcribed, and personal identifiers were replaced with a broad profession-specialty label (e.g. Infectious Disease Physician 1). Two research team members conducted the interviews: a pharmacist with experience conducting semi-structured interviews (MG) who had an existing professional relationship with some of the interviewed pharmacists, and a student-pharmacist (ND) who was

trained on the interview process and observed multiple interviews during her training, and did not have any prior interactions with the interviewees.

The initial aim was to interview at least two healthcare providers within each professional group—specialty combination (two intensive care physicians, two infectious disease physicians, etc.) for a minimum of 24 interviews, after which data saturation would be assessed [31]. Data saturation was defined *a priori* as being met when the addition of at least two new interviews did not result in the identification of any new beliefs [32]. However, it proved difficult to contact emergency medicine nurses due to stressors associated with the coronavirus disease 2019 pandemic, and the decision was made to perform the initial thematic analysis and assess for data saturation following 22 interviews, without interviewing any emergency medicine nurses [33]. Following the initial 22 interviews, three additional interviews were conducted – one with each provider group – in order to enhance the confidence in achieving thematic saturation and to reach the initial interview goal of 24 interviews.

Data analysis of interview material

NVivo Version 12 was used for transcription and data analysis [34]. A coding guideline was generated through an iterative consensus process between both interviewers (MG, ND) to categorize segments into one of the 17 domains of the TDF [35]. The coding guideline was then used for deductive analysis of all 25 transcripts. This deductive analysis process was completed independently by each reviewer to allow for assessment of agreement and reliability (Appendix 2, see online supplementary material) [23,36]. Following deductive analysis, one reviewer (MG) performed inductive analysis on the coded segments to generate belief statements which corresponded to the TDF domains [37]. The second reviewer (ND) independently reviewed and interrogated these belief statements against the deductive analysis results [23]. The inductive analysis process was conducted using 20 of the 25 interview transcripts. Two transcripts were selected at random as a hold-out sample, in addition to three transcripts from interviews conducted after the initial set of 22 interviews which were conducted intentionally to assess for data saturation. The second reviewer (ND) reviewed these five transcripts independently and found no additional beliefs that had not been identified from the initial 20 transcripts, meeting the criteria for data saturation, and no additional interviews were conducted [31,38].

Translating barriers to intervention functions, behaviour change techniques and proposed interventions

The TDF themes for the barriers identified in the interview were mapped to the intervention functions following the framework according to Steps 7 and 8 of the eight-step process described in the COM-B/BCW framework. Further, these intervention functions were translated to behaviour change techniques. Finally, based on practical experience, previous published work and consideration of realistic changes, a list of proposed implementation interventions was produced.

Table 1
Interviewee demographics and clinical background

Characteristic	Healthcare provider count, N (%) (N=25)
Sex	
Female	15 (60%)
Practice location	
USA	23 (92%)
Pennsylvania	16 (64%)
Ohio	1 (4%)
New York	2 (8%)
North Carolina	1 (4%)
Georgia	1 (4%)
Illinois	2 (8%)
Canada	1 (4%)
Europe	1 (4%)
Clinical profession	
Physician	9 (36%)
Pharmacist	9 (36%)
Nurse	7 (28%)
Clinical specialty	
General practice ^a	8 (32%)
Infectious disease/ infection prevention	6 (24%)
Emergency medicine	4 (16%)
Intensive care	7 (28%)

^a Included hospitalist physicians, family medicine physicians, hospital staff pharmacists and hospital staff nurses.

Results

In total, 25 in-depth interviews were conducted with a total of nine physicians, nine pharmacists and seven nurses. General practice included hospitalist or family medicine physicians, and hospital staff nurses and pharmacists. Almost all (23/25) interviewees were practising in the USA at the time of the interview. Interviewee background information is summarized in Table 1.

Seventy belief statements were identified across the 14 TDF domains through the inductive analysis process and are shown in Table 2. The frequency of each belief statement, as expressed by each provider group and clinical specialty, was evaluated in order to identify the most important beliefs [37]. Each belief was counted if it was expressed in the particular interview, and could only be counted once per interview. The domain of environmental context and resources generated the most unique themes (N=9), and the beliefs within this domain were the most frequently expressed overall. The domains of optimism, goals and emotion generated the fewest beliefs (N=3 each), and were the least frequently expressed domains overall.

The most frequently expressed beliefs were translated into two proposed interventions using the BCW, and this process is summarized in Table 3 [21]. The first proposed intervention is for health systems to develop and disseminate a clear policy for the evaluation of BL allergies while promoting the use of

Table II

Inductive analysis belief statements by professional group, clinical practice area and overall results

TDF domain	Belief statements by participants	Results by professional group			Results by clinical practice area				Overall results
		MD (N=9)	RPh (N=9)	RN (N=7)	ID (N=6)	EM (N=4)	IC (N=7)	GP (N=8)	Total (N=25)
Environmental context and resources	Evaluating allergies is too time consuming	4	4	3	1	2	2	6	11
	I have higher priorities in my care setting	4	4	4	2	3	3	4	12
	There are safeguards/alerts that assist me	1	1	2	2	0	1	1	4
	Patients and family are unavailable or unreliable	9	6	7	6	4	7	5	22
	Severe reactions are better remembered and documented	3	2	5	2	0	4	4	10
	Documentation of information is missing or unreliable	9	8	3	4	4	7	5	20
	I am confident in the information provided when it is available	8	6	4	4	4	6	4	18
	Additional forms of testing would be beneficial	6	3	0	3	1	2	3	9
	I refer to external resources for the evaluation process	6	2	3	2	1	4	4	11
Beliefs about capabilities	Most allergies can be determined with the proper tools and information	8	8	4	5	2	6	7	20
	I am confident in my ability to evaluate an allergy	9	9	6	5	4	7	8	24
	Evaluating allergies is not difficult	5	5	3	3	0	5	5	13
	I am confident in removing/amending allergy documentation	9	9	2	4	4	6	6	20
	I am not confident in removing/amending allergy documentation	1	0	5	2	1	1	2	6
	I am comfortable challenging erroneous allergies	2	3	0	1	3	0	1	5
Behavioural regulation	There is no policy or it is only informal	6	7	5	4	3	6	5	18
	I do not know if we have a policy	4	3	4	2	3	3	3	11
	I am not confident in the current policy/procedure	0	4	2	3	2	0	1	6
	A formal policy would be beneficial	1	5	3	3	1	3	2	9
	We do have a formal policy	1	2	3	4	0	2	0	6
	Policies do not dictate my evaluations	1	1	1	0	1	1	1	3
Beliefs about consequences	I prefer to be safe if I am unsure	5	2	4	2	1	3	5	11
	Evaluating allergies will improve patient outcomes and safety	3	2	0	1	0	0	4	5
	I have seen negative consequences from allergy evaluation/challenging	3	0	0	0	2	1	0	3
Emotion	I am worried I may hurt the patient	2	0	3	0	0	2	3	5
	This process is not stressful	9	9	5	6	4	6	7	23
	This process is stressful	1	0	2	0	0	2	1	3

(continued on next page)

Table II (continued)

TDF domain	Belief statements by participants	Results by professional group			Results by clinical practice area				Overall results
		MD (N=9)	RPh (N=9)	RN (N=7)	ID (N=6)	EM (N=4)	IC (N=7)	GP (N=8)	Total (N=25)
Goals	My goal is to provide high-quality care	4	5	3	4	1	4	3	12
	My goal is to clarify the allergy	4	5	4	2	1	5	5	13
	My goal is to treat the patient immediately	5	0	3	1	2	4	1	8
Intentions	Every allergy needs to be evaluated every time	1	3	2	1	2	1	2	6
	Allergy evaluation is very high priority in some circumstances	6	7	5	3	2	7	6	18
	The allergy does not need to be evaluated unless a penicillin is being given	7	5	4	5	3	3	5	16
	The allergy reaction is the most important factor for evaluation	9	8	3	3	4	5	8	20
	I do not evaluate well-documented allergies	0	4	1	1	1	1	2	5
	There is confusion between side effects and allergies	5	4	5	3	2	4	5	14
	I know what a true allergy is	0	3	3	2	1	1	2	6
Knowledge	There is a lack of knowledge/education on this topic	3	3	3	4	0	2	3	9
	Beta-lactams are the optimal choice for some infections	7	4	1	4	1	4	3	12
	I know which beta-lactams I can use based on side chain activity	1	7	0	3	0	2	3	8
Memory, attention and decision process	I can use alternative non-beta-lactam antibiotics (or generally other options)	5	3	1	2	2	3	2	9
	I can use alternative beta-lactam antibiotics	3	6	1	4	3	3	0	10
	I can use anything a patient has received previously	3	5	0	2	3	2	1	8
	Every patient has different needs	3	4	1	2	2	1	3	8
	It is not possible to evaluate some allergies confidently	2	3	2	1	2	3	1	7
Optimism	Most documented allergies are not true allergic reactions	3	3	1	2	2	2	1	7
	Allergies are not evaluated as often as they should be	2	3	3	1	0	2	5	8
	There is little to no oversight on allergy evaluation	1	1	2	0	2	2	0	4
Reinforcement	There is no reward or recognition for evaluating allergies	8	5	4	3	3	5	6	17
	I am unlikely to be reprimanded whether or not I evaluate allergies	7	6	3	3	3	4	6	16
	I am personally satisfied by providing good care to the patient	2	3	0	1	1	1	2	5
Skills	I am guided by what others have taught me	3	0	0	1	0	1	1	3
	I have experience evaluating allergies	4	7	5	5	1	6	4	16

Table II (continued)

TDF domain	Belief statements by participants	Results by professional group			Results by clinical practice area				Overall results
		MD (N=9)	RPh (N=9)	RN (N=7)	ID (N=6)	EM (N=4)	IC (N=7)	GP (N=8)	Total (N=25)
Social and professional role and identity	Evaluating allergies requires experience	2	1	2	2	1	1	1	5
	Navigating the EHR is an important skill	3	6	1	3	1	3	3	10
	Interviewing patients/family is an important skill	5	4	2	3	2	3	3	11
	Nurses are too busy to evaluate allergies	1	0	1	0	1	1	0	2
	Nurses are relied on for information gathering, not evaluation	1	0	5	2	0	1	3	5
	All clinicians should know how to evaluate allergies	1	1	2	2	0	2	0	4
	It is the prescriber's job to evaluate allergies	6	2	3	1	3	4	3	11
	It is everyone's job to evaluate allergies	1	6	4	2	1	4	4	11
	Pharmacists are the most equipped to evaluate allergies	5	8	5	5	2	6	5	18
	Evaluating allergies is part of my job	3	4	1	4	1	2	1	8
Social influences	I have colleagues I can consult when needed	4	4	4	4	2	5	1	12
	I am confident in my colleagues	5	4	4	5	1	3	4	13
	Evaluating allergies is a team effort	3	3	3	4	1	2	2	9
	I am prompted by others to evaluate allergies	2	1	2	1	0	1	3	5
	I rely on others to make the final decision	0	1	6	2	0	2	3	7
	Evaluating allergies is a high priority to my employer/supervisor	3	6	3	4	1	3	4	12
	Evaluating allergies is not a high priority to my employer/supervisor	3	2	2	2	0	1	4	7

TDF, Theoretical Domains Framework; MD, physicians; RPh, pharmacists; RN, nurses; ID, infectious disease/disease prevention; EM, emergency medicine; IC, intensive care; GP, general practice including hospitalist physicians, family medicine physicians, hospital staff pharmacists and hospital staff nurses; EHR, electronic health record.

pharmacists in the evaluation of drug allergies, which involves: (1) training pharmacists and technicians to act in this role; (2) environmental restricting; (3) enabling – allowing pharmacists and technicians the time and access required to perform this activity; (4) restriction; and (5) modelling – assigning responsibility of this role to pharmacists and allowing them to act as the champion for the evaluation of BL allergies where possible. The second proposed intervention is a complete rework of the EHR process for documenting and utilizing allergy information. This intervention would require: (1) enablement; and (2) environmental restricting – reworking the allergy documentation process to utilize picklists of allergy reactions instead of free text; (3) education; (4) training; and (5) modelling – the proper use of entering and modifying allergy information in the EHR to allow for clear usage of allergy information in the future.

Quotes assessed to develop proposed intervention 1: a clear policy for the evaluation of BL allergies while promoting the use of pharmacists in the evaluation of drug allergies

Many interviewees reported that evaluation of BL allergies was not guided by a clear policy as no such policy existed, and that this was an informal process that was taught to them through practice:

'I actually don't know if there's a formal hospital policy on that process, that's generally what I do [to evaluate an allergy].'
(Intensive Care Nurse 2)

There was a lack of knowledge on whether a policy for proper evaluation of BL allergies existed and overall low confidence in such policies whenever they were present:

Table III

Belief statement intervention recommendation translation summary using the Behaviour Change Wheel function categories

TDF domain	Relevant belief statement(s)	Behaviour Change Wheel intervention function categories	Behaviour change techniques	Proposed intervention
Behavioural regulation	• There is no policy or it is only informal	• Training • Environmental restructuring	• Demonstration of the behaviour • Social support • Restructuring the physical environment	Develop and disseminate a policy for the evaluation of beta-lactam allergies while promoting the use of pharmacists in the evaluation of drug allergies Rework of the EHR process for documenting and accessing allergy information
Social influences	• I do not know if we have a policy • I have colleagues I can consult when needed	• Enablement		
Social and professional role and identity	• I am confident in my colleagues • Pharmacists are the most equipped to evaluate allergies	• Modelling • Restriction		
Beliefs about capabilities	• Most allergies can be determined with the proper tools and information • I am confident in my ability to evaluate an allergy • I am confident in removing/amending allergy documentation	• Enablement	• Restructuring the physical environment • Review the behaviour goal • Review the outcome goal • Adding objects to the environment • Self-monitoring of behaviour • Instruction on how to perform a behaviour	
Knowledge	• There is confusion between side effects and allergies • Beta-lactams are the optimal choice for some infections	• Education		
Intentions	• The allergy reaction is the most important factor for evaluation • The allergy does not need to be evaluated unless a penicillin is being given	• Modelling		
Memory, attention and decision process	• I can use alternative agents (both beta-lactams and non-beta-lactams) • I can use anything a patient has tolerated previously	• Training • Environmental restructuring • Enablement		
Environmental context and resources	• Patients and family are unavailable or unreliable • Documentation of information is missing or unreliable • I am confident in the information provided when it is available	• Environmental restructuring • Enablement		

TDF, Theoretical Domains Framework; EHR, electronic health record.

'I wouldn't say we have an official, in our organization — policy and procedure. In fact, we don't have an official policy or procedure.' (General Practice Physician 2)

'OK, so from a pharmacist standpoint ... Yeah, I'm not that confident, especially considering our current model in our hospital is more of a centralized staff pharmacies model, so a lot of times there's not a direct interaction with the patients.' (Emergency Department Pharmacist 1)

In contrast to the lack of confidence in formal policies and procedures, interviewees reported a high degree of confidence in their colleagues to assess BL allergies properly, particularly infectious disease specialists:

'But with other infectious diseases pharmacists, I think they're all very good at assessing allergies and investigating further.' (Infectious Diseases Pharmacist 2)

In addition to high confidence in their colleagues, there was a high level of accessibility reported, and many interviewees reported they had colleagues that they could consult to assist with the evaluation of BL allergies when they did not have the expertise or the resources required for a particular evaluation:

'[I contact] The physician and the pharmacist. I'll call the physician and then a lot of the times, the physician will even ask you to call the pharmacist and see what they think about the dosage difference and things like that. So, I do call both of them.' (General Practice Nurse 1)

In response to being asked 'Who is the most equipped to evaluate beta-lactam allergies?', interviewees overwhelmingly identified pharmacists for their background as medication experts, which many felt extended to drug allergies. More than half of all interviewees expressed this belief across each professional group, including seven of eight interviewed pharmacists:

'I do think that the pharmacists are better than most of us about doing these things because they deal with meds. They understand the cross-reactivity between different meds and everything else. They're much better at looking at the med histories than most of us are.' (Emergency Medicine Physician 1)

Quotes assessed to develop proposed intervention 2: a rework of the EHR process for documenting and utilizing allergy information

Frustration with the lack of consistency and reliability of allergy information contained within the EHR was a common topic that arose throughout the interviews. There was a belief that the critical component of the allergy reaction was not well documented or contained vague information:

'Allergies are listed in the EHR and it comes with a descriptor of what the allergy actually is. Many times, those are very short statements or a couple of words. One might just say rash. And so, trying to determine what that actually means is sometimes challenging.' (Intensive Care Nurse 2)

This frustration was amplified because the allergy reaction was consistently identified as the most important component in determining the legitimacy of an allergy:

'The first thing I usually do when I come across the beta-lactam allergy is look to see the specific reaction that is listed.' (Intensive Care Pharmacist 2)

Patients and family, when available as a source of information, were not thought to be reliable reporters of allergy information, and healthcare providers often felt that patients and family could not distinguish between a true immune-mediated allergy or a routine adverse effect:

'As I said, patients are notoriously bad, they, you know, over half of them have been told by their mother that they just can't have penicillin and nobody knows why. And if you were to go ask that parent and take that one step further, then maybe they won't remember either. So, I don't think patients are very reliable at giving that information.' (Infectious Disease Physician 1)

In contrast, interviewees did have a relatively high degree of confidence in allergy information when it was fully available, were confident in their own abilities to evaluate BL allergies, and felt that most BL allergies could be evaluated properly when proper information was provided:

'If whoever is investigating the allergy gets the information that they need. I think that it certainly can be determined by asking the right questions.' (General Practice Pharmacist 1)

A lack of the required time, as well as the ability to treat the patient adequately with alternative agents or using agents that the patient has tolerated previously, was often cited as a reason for not thoroughly evaluating or challenging allergies:

'OK, we'll give you [clindamycin] for your dental infection. Maybe not the best choice. But most people aren't going to get to the bottom of it either, it's just too labor intensive.' (Emergency Department Physician 1)

'If I clearly see that they've received similar antibiotics safely in the past then I do very little to no further investigation at that point.' (Intensive Care Pharmacist 2)

BL antibiotics were recognized as the optimal option for many situations despite the challenges expressed in utilizing them in the setting of an allergy:

'And again, it's all about the antimicrobial stewardship. For example, where I practise is a region that has a high syphilis rate. So being able to give penicillin opposed to doxycycline, it's important for treatment.' (Infectious Disease Physician 2)

Discussion

Twenty-five front-line healthcare providers were interviewed to obtain their perspectives about evaluating the legitimacy of BL allergies in practice. Some of the more prominent barriers to the evaluation of allergies were focused on unclear or non-existent protocols for the evaluation of allergies, a lack of time to evaluate BL allergies relative to higher-priority tasks, and frustration with missing information in allergy documentation, particularly regarding the allergy reaction. Using the BCW framework for analysis, these beliefs were then translated into two recommended implementation interventions: first, health systems should create and disseminate clear policies for the proper evaluation of BL allergies while promoting the use of pharmacists in the evaluation of allergies when possible; and second, a complete rework of the EHR process for documenting and utilizing allergy information.

It is strongly recommended that health systems should adopt and widely disseminate clear policies about the time, place and person who is primarily responsible for evaluating drug allergies, and the results of this study highlight

pharmacists as ideal candidates for managing the evaluation of allergies. Pharmacist-led and pharmacist-technician-assisted models for BL allergy have a proven track record of success, and can be customized to meet the unique needs and resources of each particular health setting [39–42]. Once provided with proper training from allergists and infectious disease staff, pharmacists have been shown to be able to properly risk-stratify patients and provide referrals for additional services such as skin testing or oral amoxicillin challenges, but are also highly accurate when using structured clinical history alone if these additional resources are unavailable [40,41,43]. Evaluation of BL allergies is highly cost effective and can result in reduced rates of adverse drug events [44,45]. The interview results in the present study revealed a lack of consistency about who was thought to be responsible for the evaluation of allergies, even among healthcare providers within the same health system. Both physicians and nurses reported a high degree of confidence in pharmacists in this role, and supported the use of pharmacists to evaluate drug allergies to alleviate the time and priority-related concerns associated with this task.

Allergy documentation in EHRs is known to contain inconsistencies, and differentiates poorly between drug intolerances and true immune-mediated allergies, leading to difficulty in accurate interpretation of allergy data and a high rate of allergy alerts being bypassed [46–49]. The interview results from this study mirrored these allergy documentation issues, and strongly support a rework of the EHR process for documenting and utilizing allergy information. Most importantly, an EHR rework should emphasize the importance of reporting as much information regarding the allergy reaction as possible, as this was the single most important factor reported for determining allergy severity and the need to alter prescribing practices. This could be accomplished through the use of picklists instead of free-text allergy entries [46]. Additionally, allergy decision support alerts systems should be redesigned to assist in recognizing previous patterns of tolerated antibiotic use which conflict with documented allergy information. A recent work group by the American Academy of Allergy, Asthma and Immunology has recommended similar changes to allergy documentation [46].

This study has some limitations. First, the authors sought to interview a diverse group of clinicians and specialties to capture a wide picture of the beliefs and practices for the evaluation of BL allergies, but in doing so, it was only possible to interview an average of two clinicians for each provider-specialty group, and it was not possible to interview nurses working in emergency medicine. Also, the interview group may be subject to response bias due to healthcare providers with the most knowledge on BL allergies being those most inclined to agree to be interviewed on the topic. The inductive and deductive analysis process inherently requires a degree of subjective interpretation, and although an attempt was made to reduce this using a secondary reviewer and a standardized framework, utilization of other methods such as triangulation or member checking could have been employed to further reduce subjectivity. Interviewees were aware that the study team consisted of pharmacists through the initial enrolment e-mail, which may have led to more favourable views on the abilities of pharmacists. Finally, the BCW and TDF were used for their ability to evaluate both individual-level and organizational-level constructs, instead of other frameworks

such as the Consolidated Framework for Implementation Research which excels at evaluating organizational level interventions, because this study focused on individual practitioner-level barriers that were being experienced in practice [50,51].

In conclusion, there has been a substantial increase in the understanding of erroneous BL allergies, and an increase in resources available to support accurate allergy evaluation and delabelling. It is important that interventions that utilize these resources are tailored specifically to address the current barriers in practice. The results of this study highlight policy and EHR-related concerns reported by front-line healthcare providers that should be considered by health systems which plan to implement interventions to delabel erroneous BL allergies and support improvements in the drug allergy documentation process.

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Conflict of interest statement

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhin.2023.07.024>.

References

- [1] Bush K, Bradford PA. β -lactams and β -lactamase inhibitors: an overview. *Cold Spring Harb Perspect Med* 2016;6:a025247.
- [2] Zhou L, Dhopeswarkar N, Blumenthal KG, Goss F, Topaz M, Slight SP, et al. Drug allergies documented in electronic health records of a large healthcare system. *Allergy* 2016;71:1305–13.
- [3] Macy E. Penicillin and beta-lactam allergy: epidemiology and diagnosis. *Curr Allergy Asthma Rep* 2014;14:476.
- [4] Blumenthal KG, Oreskovic NM, Fu X, Shebl FM, Mancini CM, Maniates JM, et al. High-cost, high-need patients: the impact of reported penicillin allergy. *Am J Manag Care* 2020;26:154–61.
- [5] Su T, Broekhuizen BDL, Verheij TJM, Rockmann H. The impact of penicillin allergy labels on antibiotic and health care use in primary care: a retrospective cohort study. *Clin Transl Allergy* 2017;7:18.
- [6] Charneski L, Deshpande G, Smith SW. Impact of an antimicrobial allergy label in the medical record on clinical outcomes in hospitalized patients. *Pharmacotherapy* 2011;31:742–7.
- [7] Ness RA, Bennett JG, Elliott WV, Gillion AR, Pattanaik DN. Impact of β -lactam allergies on antimicrobial selection in an outpatient setting. *South Med J* 2019;112:591–7.
- [8] Trubiano JA, Adkinson NF, Phillips EJ. Penicillin allergy is not necessarily forever. *JAMA* 2017;318:82–3.
- [9] Sacco KA, Bates A, Brigham TJ, Imam JS, Burton MC. Clinical outcomes following inpatient penicillin allergy testing: a systematic review and meta-analysis. *Allergy* 2017;72:1288–96.
- [10] Shenoy ES, Macy E, Rowe T, Blumenthal KG. Evaluation and management of penicillin allergy: a review. *JAMA* 2019;321:188–99.

- [11] Trubiano JA, Vogrin S, Chua KYL, Bourke J, Yun J, Douglas A, et al. Development and validation of a penicillin allergy clinical decision rule. *JAMA Intern Med* 2020;180:745–52.
- [12] Piotin A, Godet J, Trubiano JA, Grandbastien M, Guénard-Bilbault L, de Blay F, et al. Predictive factors of amoxicillin immediate hypersensitivity and validation of PEN-FAST clinical decision rule. *Ann Allergy Asthma Immunol* 2022;128:27–32.
- [13] Mohamed OE, Beck S, Huissoon A, Melchior C, Heslegrave J, Baretto R, et al. A retrospective critical analysis and risk stratification of penicillin allergy delabeling in a UK specialist regional allergy service. *J Allergy Clin Immunol Pract* 2019;7:251–8.
- [14] Brockow K, Wurpts G, Trautmann A. Patients with questionable penicillin (beta-lactam) allergy: causes and solutions. *Allergol Select* 2022;6:33–41.
- [15] Anstey KM, Tsao L, Otani IM. Drug allergy delabeling programs: recent strategies and targeted populations. *Clin Rev Allergy Immunol* 2022;62:484–504.
- [16] Blumenthal KG, Acker WW, Li Y, Holtzman NS, Zhou L. Allergy entry and deletion in the electronic health record. *Ann Allergy Asthma Immunol* 2017;118:380–1.
- [17] Collins K, Rueter K, Lucas M, Sommerfield D, Sommerfield A, Khan N, et al. Penicillin allergy SHACK: survey of hospital and community knowledge. *J Paediatr Child Health* 2022;58:1414–9.
- [18] Pettett BJ, Eskildsen SM, Huang KX, Ostrum RF. Despite the safety of preoperative cefazolin for patients with non-anaphylactic penicillin allergy, 20% of practitioners avoid its use. *Orthopedics* 2019;42:e437–42.
- [19] Powell N, Wilcock M, Roberts N, Sandoe J, Tonkin-Crine S. Focus group study exploring the issues and the solutions to incorrect penicillin allergy-labelled patients: an antibiotic stewardship patient safety initiative. *Eur J Hosp Pharm* 2021;28:71–5.
- [20] Wanat M, Anthierens S, Butler CC, Savic L, Savic S, Pavitt SH, et al. Patient and primary care physician perceptions of penicillin allergy testing and subsequent use of penicillin-containing antibiotics: a qualitative study. *J Allergy Clin Immunol Pract* 2019;7:1888–1893.e1.
- [21] Michie S, van Stralen MM, West R. The behaviour change wheel: a new method for characterising and designing behaviour change interventions. *Implement Sci* 2011;6:42.
- [22] O'Brien BC, Harris IB, Beckman TJ, Reed DA, Cook DA. Standards for reporting qualitative research: a synthesis of recommendations. *Acad Med* 2014;89:1245–51.
- [23] Atkins L, Francis J, Islam R, O'Connor D, Patey A, Ivers N, et al. A guide to using the Theoretical Domains Framework of behaviour change to investigate implementation problems. *Implement Sci* 2017;12:77.
- [24] Ojo SO, Bailey DP, Brierley ML, Hewson DJ, Chater AM. Breaking barriers: using the behavior change wheel to develop a tailored intervention to overcome workplace inhibitors to breaking up sitting time. *BMC Public Health* 2019;19:1126.
- [25] Coyne IT. Sampling in qualitative research. Purposeful and theoretical sampling; merging or clear boundaries? *J Adv Nurs* 1997;26:623–30.
- [26] Streeton R, Cooke M, Campbell J. Researching the researchers: using a snowballing technique. *Nurse Res* 2004;12:35–46.
- [27] Kallio H, Pietilä AM, Johnson M, Kangasniemi M. Systematic methodological review: developing a framework for a qualitative semi-structured interview guide. *J Adv Nurs* 2016;72:2954–65.
- [28] Baumbusch J. Semi-structured interviewing in practice-close research. *J Spec Pediatr Nurs* 2010;15:255–8.
- [29] Cridland EK, Jones SC, Caputi P, Magee CA. Qualitative research with families living with autism spectrum disorder: recommendations for conducting semistructured interviews. *J Intellect Dev Disabil* 2015;40:78–91.
- [30] Archibald MM, Ambagtsheer RC, Casey MG, Lawless M. Using Zoom videoconferencing for qualitative data collection: perceptions and experiences of researchers and participants. *Int J Qualitat Meth* 2019;18:1609406919874596.
- [31] Morse JM. Determining sample size. Thousand Oaks, CA: Sage; 2000. p. 3–5.
- [32] Francis JJ, Johnston M, Robertson C, Glidewell L, Entwistle V, Eccles MP, et al. What is an adequate sample size? Operationalising data saturation for theory-based interview studies. *Psychol Health* 2010;25:1229–45.
- [33] Cornish S, Klim S, Kelly AM. Is COVID-19 the straw that broke the back of the emergency nursing workforce? *Emerg Med Australas* 2021;33:1095–9.
- [34] NVivo. Burlington, MA: QST International Pty Ltd; 2020.
- [35] Hsieh HF, Shannon SE. Three approaches to qualitative content analysis. *Qual Health Res* 2005;15:1277–88.
- [36] McHugh ML. Interrater reliability: the kappa statistic. *Biochem Med (Zagreb)* 2012;22:276–82.
- [37] Islam R, Timmouth AT, Francis JJ, Brehaut JC, Born J, Stockton C, et al. A cross-country comparison of intensive care physicians' beliefs about their transfusion behaviour: a qualitative study using the Theoretical Domains Framework. *Implement Sci* 2012;7:93.
- [38] Morse JM. Analytic strategies and sample size. *Qualitat Health Res* 2015;25:1317–8.
- [39] Bhogal R, Hussain A, Balaji A, Bermingham WH, Marriott JF, Krishna MT. The role of a clinical pharmacist in spurious penicillin allergy: a narrative review. *Int J Clin Pharm* 2021;43:461–75.
- [40] Turner NA, Wrenn R, Sarubbi C, Kleris R, Lugar PL, Radojicic C, et al. Evaluation of a pharmacist-led penicillin allergy assessment program and allergy delabeling in a tertiary care hospital. *JAMA Netw Open* 2021;4:e219820.
- [41] Gugkaeva Z, Crago JS, Yasnogorodsky M. Next step in antibiotic stewardship: pharmacist-provided penicillin allergy testing. *J Clin Pharm Ther* 2017;42:509–12.
- [42] Gaudreau S, Bourque G, Côté K, Nutu C, Beauchesne MF, Longpré AA, et al. Resources assessment for penicillin allergy testing performed by pharmacists at the patient's bedside. *Ann Pharmacother* 2021;55:1355–62.
- [43] Ham Y, Sukerman ES, Lewis 2nd JS, Tucker KJ, Yu DL, Joshi SR. Safety and efficacy of direct two-step penicillin challenges with an inpatient pharmacist-driven allergy evaluation. *Allergy Asthma Proc* 2021;42:153–9.
- [44] Blumenthal KG, Shenoy ES, Wolfson AR, Berkowitz DN, Carballo VA, Balekian DS, et al. Addressing inpatient beta-lactam allergies: a multihospital implementation. *J Allergy Clin Immunol Pract* 2017;5:616–625.e7.
- [45] Bosma BE, van den Bemt P, Melief P, van Bommel J, Tan SS, Hunfeld NGM. Pharmacist interventions during patient rounds in two intensive care units: clinical and financial impact. *Neth J Med* 2018;76:115–24.
- [46] Guyer AC, Macy E, White AA, Kuruvilla ME, Robison RG, Kumar S, et al. Allergy electronic health record documentation: a 2022 Work Group Report of the AAAAI Adverse Reactions to Drugs, Biologicals, and Latex Committee. *J Allergy Clin Immunol Pract* 2022;10:2854–67.
- [47] Blumenthal KG, Park MA, Macy EM. Redesigning the allergy module of the electronic health record. *Ann Allergy Asthma Immunol* 2016;117:126–31.
- [48] Genco EK, Forster JE, Flaten H, Goss F, Heard KJ, Hoppe J, et al. Clinically inconsequential alerts: the characteristics of opioid drug alerts and their utility in preventing adverse drug events in the emergency department. *Ann Emerg Med* 2016;67:240–8.e3.
- [49] Topaz M, Seger DL, Slight SP, Goss F, Lai K, Wickner PG, et al. Rising drug allergy alert overrides in electronic health records: an observational retrospective study of a decade of experience. *J Am Med Inform Assoc* 2016;23:601–8.

- [50] Damschroder LJ, Aron DC, Keith RE, Kirsh SR, Alexander JA, Lowery JC. Fostering implementation of health services research findings into practice: a consolidated framework for advancing implementation science. *Implement Sci* 2009;4:50.
- [51] Birken SA, Powell BJ, Presseau J, Kirk MA, Lorencatto F, Gould NJ, et al. Combined use of the Consolidated Framework for Implementation Research (CFIR) and the Theoretical Domains Framework (TDF): a systematic review. *Implement Sci* 2017;12:2.